Chapter 4

Financial Statements and Business Basics

Financial statements are the language through which firms communicate their economic position to investors, lenders, regulators, and managers. They are structured summaries of how a business creates value, uses capital, and manages risk, not mere accounting artifacts. For an investor, analyst, trader, or data scientist, the ability to read them is indispensable: a model may forecast returns or defaults, but it is useless without the business context behind the numbers.

This chapter introduces the three core financial statements: the balance sheet, the income statement, and the cash flow statement. It explains how they fit together, why they differ, and how they support reasoning about profitability, liquidity, solvency, and value creation. The goal is not to turn the reader into an accountant but to build the intuition needed to analyze firms in a way that is financially sound and quantitatively useful.

A fictional manufacturer, Meridian Fabrication Inc., runs through the chapter as its central worked example, so that every ratio is tied to actual numbers. Its balance sheet, income statement, and cash flow statement are built up section by section, then mined for liquidity, leverage, profitability, and efficiency ratios (Section 4.3.5), an Altman's Z-score distress assessment, and a common-size restatement that makes its cost structure comparable across firm sizes. A second company, Solstice Robotics Inc. (Section 4.4.1), is introduced for comparison, so that a ratio's meaning becomes clear by contrast: the same debt-to-equity ratio can signal prudence at one firm and fragility at another.

The ratio toolkit built here is the one that Chapter 9's credit scoring and Chapter 6's classification models later formalize statistically. Section 4.5.1 links it to Chapter 5's valuation models, since a discounted-cash-flow model is only as trustworthy as the statement forecasts feeding it. The accounting-quality cautions raised late in the chapter (reported numbers reflect choices and estimates, not just facts) apply to every later chapter that treats statement figures as clean model inputs.

4.1 Why Statements Matter, and a Running Example

Accounting is the language a firm uses to describe itself, and like any language it can be used carefully or evasively. This part explains why the statements repay close reading, and introduces the company whose numbers the rest of the chapter works through.

4.1.1 Why Financial Statements Matter

A firm is a system for converting resources into cash flows. It buys inputs, hires labor, invests in equipment, and sells products or services. In the process, it makes decisions about financing, operations, and strategy, and financial statements summarize these decisions. They let stakeholders ask essential questions: How much debt does the firm carry? Is it profitable? Is it generating enough cash to fund operations? What portion of the business is financed by owners versus lenders? Is growth coming from operating improvement or from financial engineering?

Wulin.Suo@Queensu.ca

Finance is about the economic reality of the firm as well as price changes. A stock can look attractive because its price rises, but if the company is consuming cash and taking on excessive leverage, the apparent success may be fragile. Conversely, a firm with modest headline earnings may still be valuable if it produces durable cash flows and has strong competitive advantages. Financial statements help separate signal from noise.

The statements are analyzed together. The balance sheet shows the firm's position at a point in time, the income statement its performance over a period, and the cash flow statement the movement of cash; together they give a more complete picture than any one report. Figure 4.1 sketches how the three statements relate to one another.

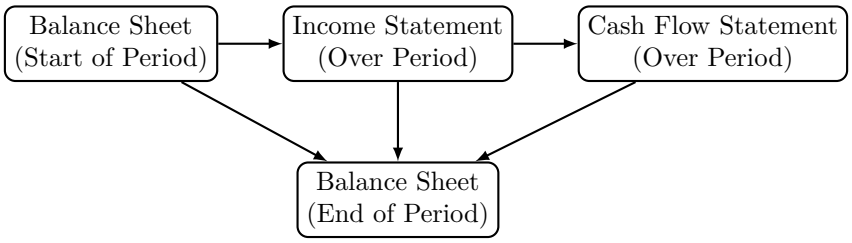


Figure 4.1: The three statements are linked: the income statement and cash flow statement explain the change in the balance sheet from the start to the end of the period

4.1.2 A Running Example: Meridian Fabrication Inc.

Meridian Fabrication Inc. is a small, fictional industrial parts manufacturer. All figures are in millions of dollars. Table 4.1 shows a simplified balance sheet at the end of two consecutive fiscal years, Table 4.2 shows the income statement for the most recent year, and Table 4.3 shows the cash flow statement for the same year. These numbers are used throughout the chapter to illustrate ratios, forecasting, and the limits of reported figures.

	Year 1	Year 2
Cash and equivalents	12.0	9.0
Accounts receivable	18.0	24.0
Inventory	22.0	30.0
Total current assets	**52.0**	**63.0**
Property, plant, and equipment (net)	60.0	66.0
Total assets	**112.0**	**129.0**
Accounts payable	14.0	17.0
Short-term debt	6.0	8.0
Total current liabilities	**20.0**	**25.0**
Long-term debt	40.0	50.0
Total liabilities	**60.0**	**75.0**
Shareholders' equity	52.0	54.0
Total liabilities and equity	**112.0**	**129.0**

Table 4.1: Meridian Fabrication Inc. – Balance Sheet ($ millions)

A first observation: Meridian's net income of $9.0 million looks respectable, but cash flow from operations is only $6.0 million, and after $14.0 million of capital expenditures the firm burned cash and had to raise $12.0 million of new debt to keep its cash balance from falling further. Financial statement analysis is designed to surface this kind of gap between accounting profit and cash generation, and it recurs throughout the chapter.

Wulin.Suo@Queensu.ca

Revenue	150.0
Cost of goods sold	96.0
Gross profit	**54.0**
Selling, general, and administrative expense	30.0
Depreciation and amortization	8.0
Operating income (EBIT)	**16.0**
Interest expense	4.0
Pretax income	**12.0**
Taxes (25%)	3.0
Net income	**9.0**

Table 4.2: Meridian Fabrication Inc. – Income Statement, Year 2 ($ millions)

Net income	9.0
+ Depreciation and amortization	8.0
− Increase in accounts receivable	6.0
− Increase in inventory	8.0
+ Increase in accounts payable	3.0
Cash flow from operations	**6.0**
Capital expenditures	14.0
Cash flow from investing	**−14.0**
Net new debt issued	12.0
Dividends paid	7.0
Cash flow from financing	**5.0**
Net change in cash	**−3.0**

Table 4.3: Meridian Fabrication Inc. – Cash Flow Statement, Year 2 ($ millions)

4.2 The Three Statements

A firm reports itself through three linked documents: what it owns and owes, what it earned, and what cash moved. This part takes each in turn and shows how they articulate with one another.

4.2.1 The Balance Sheet

The balance sheet is a snapshot of a firm's resources and claims at a specific date. It is governed by a simple accounting identity:

$$\text{Assets} = \text{Liabilities} + \text{Equity}. \tag{4.1}$$

Assets are what the firm owns or controls. They include cash, receivables, inventories, property, plant and equipment, intangible assets, and other resources expected to generate future benefit. Liabilities are obligations the firm owes to others, such as loans, accounts payable, or taxes payable. Equity is the residual interest of owners after liabilities are subtracted from assets. In Table 4.1, this identity holds in both years: $112.0 million of assets equal $60.0 million of liabilities plus $52.0 million of equity in Year 1, and $129.0 million equal $75.0 million plus $54.0 million in Year 2.

The balance sheet reveals the firm's financial structure. A firm with large cash reserves and modest debt is often more resilient than one with high leverage and weak liquidity. Significant intangible assets may look valuable but be hard to liquidate in distress, and high inventory creates operational risk if demand falls unexpectedly. Meridian's inventory grew from $22.0 million to $30.0 million, a 36% increase, and its receivables from $18.0 million to $24.0 million, a 33% increase, while its revenue grew only 11%, from $135.0 million in Year 1 to the $150.0 million of Year 2. Comparing the growth of inventory and receivables with

Wulin.Suo@Queensu.ca

the growth of sales is one of the simplest early-warning checks an analyst performs, and here it flags the working-capital buildup that the cash flow statement shows consuming Meridian's cash.

The balance sheet can be analyzed for liquidity (the ability to meet short-term obligations), capital structure (the mix of debt and equity financing), and asset quality (whether reported assets are likely to generate value). Analysts commonly compute current assets divided by current liabilities, debt divided by equity, and long-term debt relative to total assets as a first pass; Section 4.3.5 computes them for Meridian.

4.2.2 The Income Statement

The income statement summarizes revenues, expenses, and profit over a period such as a quarter or a year. It begins with revenue, which is the amount earned from selling goods or services. From revenue, the firm subtracts costs of goods sold and operating expenses to arrive at operating profit, often called earnings before interest and taxes (EBIT). Further deductions for interest and taxes lead to net income, which is often the headline figure quoted in the media.

For Meridian, revenue of \$150.0 million minus cost of goods sold of \$96.0 million yields gross profit of \$54.0 million, a gross margin of

$$\text{Gross margin} = \frac{54.0}{150.0} = 36.0\%.$$

After subtracting SG&A (\$30.0 million) and depreciation (\$8.0 million), operating income is \$16.0 million, an operating margin of

$$\text{Operating margin} = \frac{16.0}{150.0} = 10.7\%.$$

After \$4.0 million of interest expense and \$3.0 million of taxes (a 25% rate applied to \$12.0 million of pretax income), net income is \$9.0 million, a net margin of $9.0/150.0 = 6.0\%$.

Net income is widely reported but should not be read mechanically. Accounting profit is influenced by depreciation, amortization, one-time gains or losses, and estimates of reserves and provisions, all of which affect apparent profitability. Analysts therefore also look at operating income, adjusted earnings, and cash-based measures of performance.

Profitability is central to valuation, because investors care whether the firm can earn returns above its cost of capital. A company may report growing sales, but if margins are declining and competition is intensifying, future value may be under pressure. Profitability analysis therefore examines gross, operating, and net margins and their trend over time, not just their level in one year.

One further measure appears so often in valuation practice that it belongs alongside these margins: EBITDA, earnings before interest, taxes, depreciation, and amortization, computed by adding depreciation and amortization back to operating income. For Meridian, EBITDA $= 16.0 + 8.0 =$ \$24.0 million, an EBITDA margin of $24.0/150.0 = 16.0\%$. EBITDA is popular because it strips out depreciation and amortization, non-cash charges that vary across firms with accounting choices and asset age, making operating performance easier to compare across companies with different capital structures and depreciation schedules. It is not a substitute for free cash flow: it ignores capital expenditures and working-capital changes, which Meridian's numbers show can turn a profitable-looking income statement into a cash-consuming year. As a cross-sectional comparison tool, and as the denominator of the EV/EBITDA multiple used in Chapter 5, it is worth computing routinely alongside the three margins.

4.2.3 The Cash Flow Statement

The cash flow statement is often the most informative statement about financial health. Cash keeps a business alive, and a company can report profits and still face liquidity problems if its cash is tied up in inventory, receivables, or capital spending. The statement shows how cash is generated and used in three activities: operating, investing, and financing.

Operating cash flow reflects cash generated from the core business. It starts from net income and adjusts for non-cash items (such as adding back depreciation) and for changes in working-capital accounts. Investing cash flow reflects purchases and sales of long-term assets, primarily capital expenditures. Financing cash flow reflects borrowing, repayments, dividends, and equity issuance. Table 4.3 shows this decomposition for Meridian: operating cash flow of \$6.0 million, investing cash flow of $-$\$14.0 million, and financing cash flow of

Wulin.Suo@Queensu.ca

$5.0 million, which together produce a net decrease in cash of $3.0 million, consistent with the cash balance falling from $12.0 million to $9.0 million on the balance sheet.

A particularly important concept is free cash flow, typically defined as

$$\text{Free cash flow} = \text{Cash flow from operations} - \text{Capital expenditures}. \tag{4.2}$$

For Meridian, free cash flow in Year 2 is $6.0 - 14.0 = -\$8.0$ million. This number tells a sharper story than net income: despite a $9.0 million profit, Meridian generated negative $8.0 million of free cash flow and had to borrow to cover the shortfall. Free cash flow matters for valuation because firms that generate it consistently can fund dividends, debt repayment, or reinvestment without external financing. Many valuation frameworks use free cash flows rather than accounting earnings, which non-cash charges and discretionary accounting choices can distort.

4.3 Ratio Analysis

Raw statement figures are not comparable across firms or across years; ratios are what make them so. This part builds the standard families of ratio, from short-term liquidity through leverage to the returns that reveal whether a firm creates value at all.

4.3.1 Working Capital and Liquidity

Liquidity, the ability to meet short-term obligations, is one of the clearest indicators of short-run financial health. A business that cannot pay suppliers, employees, or creditors on time may face disruption even if its long-term outlook is strong, so analysts watch current assets, current liabilities, and the cycle of converting inventory and receivables into cash.

Working capital is defined as

$$\text{Working capital} = \text{Current assets} - \text{Current liabilities}.$$

For Meridian, working capital rose from $52.0 - 20.0 = \$32.0$ million in Year 1 to $63.0 - 25.0 = \$38.0$ million in Year 2. On its own this looks like a healthy increase, but the cash flow statement shows that most of it is cash trapped in receivables and inventory rather than sitting in the bank. Too little working capital can cause distress, while too much may indicate inefficiency or, as in Meridian's case, a buildup that has not yet been converted back into cash.

Working capital is actively managed, not just measured after the fact. A treasury team can accelerate collections with early-payment discounts, negotiate longer payment terms with suppliers, or use just-in-time ordering to reduce the cash tied up in stock. Each lever maps onto one component of the cash conversion cycle computed below: shortening DSO (days sales outstanding) through faster collections, lengthening DPO (days payable outstanding) through supplier negotiation, or shortening DIO (days inventory outstanding) through leaner inventory management. Working capital financing, such as a revolving credit facility sized to bridge the gap between paying suppliers and collecting from customers, is common because this gap, as Meridian's cash flow statement shows, can consume substantial cash even for a healthy, growing business.

The cash conversion cycle is a useful way to quantify this. It is commonly expressed as

$$\text{Cash conversion cycle} = \text{DIO} + \text{DSO} - \text{DPO}, \tag{4.3}$$

where DIO measures how long inventory sits before being sold, DSO measures how long it takes to collect receivables after a sale, and DPO measures how long the firm takes to pay its own suppliers. COGS, introduced earlier as the amount subtracted from revenue to reach gross profit, is the direct cost (materials, direct labor, and manufacturing overhead) of producing what the firm sold during the period.

DIO and DPO divide by COGS rather than revenue because inventory and accounts payable are carried on the balance sheet at cost, not at the higher price a customer pays, so a cost-basis denominator keeps numerator and denominator on the same footing. DSO divides accounts receivable, recorded at the amount billed to the customer, by revenue, the corresponding billed-price figure; swapping the denominators would mix cost-basis and price-basis quantities in one ratio. Using Year 2 year-end balances and a 365-day year

Wulin.Suo@Queensu.ca

(analysts often average the opening and closing balances instead, which for a firm whose working capital is growing as fast as Meridian's gives a noticeably shorter cycle),

$$\begin{aligned}
\text{DIO} &= \frac{\text{Inventory}}{\text{COGS}} \times 365 = \frac{30.0}{96.0} \times 365 \approx 114 \text{ days},\\
\text{DSO} &= \frac{\text{Accounts receivable}}{\text{Revenue}} \times 365 = \frac{24.0}{150.0} \times 365 \approx 58 \text{ days},\\
\text{DPO} &= \frac{\text{Accounts payable}}{\text{COGS}} \times 365 = \frac{17.0}{96.0} \times 365 \approx 65 \text{ days}.
\end{aligned}$$

Rounding each component to the nearest day before summing gives $114 + 58 - 65 = 107$ days, but computing the cycle from the unrounded day-counts gives $114.06 + 58.40 - 64.64 \approx 107.8$, which rounds to 108 days. The one-day gap is a rounding artifact from summing already-rounded figures, a reminder to carry full precision through a multi-step calculation and round only the final answer. Either way, Meridian's cash is tied up in the operating cycle for roughly three and a half months. A shorter cycle is generally better, since the firm recycles cash faster and relies less on external financing, and the cycle is the kind of quantity a data-driven monitoring system can track continuously across many firms or product lines.

4.3.2 Leverage and Solvency

Leverage refers to the extent to which a firm uses debt financing. Debt is often cheaper than equity and its interest may be tax-deductible, but leverage also increases financial risk: if cash flows are volatile, debt can become burdensome and force restructuring or bankruptcy.

Solvency is the ability of a firm to meet its long-term obligations. Analysts examine leverage ratios such as debt-to-equity,

$$\text{Debt-to-equity} = \frac{\text{Total debt}}{\text{Shareholders' equity}},$$

and interest coverage,

$$\text{Interest coverage} = \frac{\text{EBIT}}{\text{Interest expense}}.$$

For Meridian in Year 2, total debt is short-term debt plus long-term debt, $8.0 + 50.0 = \$58.0$ million, against equity of \$54.0 million, giving a debt-to-equity ratio of $58.0/54.0 \approx 1.07$: the firm is financed with slightly more debt than equity. Interest coverage is $16.0/4.0 = 4.0$, meaning operating income covers interest expense four times over. A coverage ratio of 4.0 is generally considered comfortable, but the trend matters: if debt keeps growing faster than operating income, coverage will fall and the cushion against a bad year will shrink.

Leverage is not inherently good or bad; it depends on the business model, cash-flow stability, and access to financing. A utility with stable revenues can carry significant debt, while a startup in a volatile industry may support little. The question is whether leverage amplifies the firm's strengths or its weaknesses, and for Meridian, needing new debt to fund a cash shortfall driven by working-capital growth is a signal worth investigating.

4.3.3 Sector-Specific Considerations in Ratio Analysis

Ratios must be interpreted in context, and a large-scale, cross-sectional model cannot apply one set of thresholds to every firm it scores. A few contrasting industries show what that context looks like.

Banks and other financial institutions are the most extreme example. A bank's balance sheet consists mostly of financial assets (loans) funded mostly by financial liabilities (deposits). A debt-to-equity ratio that would signal extreme distress for an industrial firm like Meridian, often 8 or 10 times equity, is routine for a bank, whose business model is to hold a leveraged portfolio of loans. Standard liquidity ratios such as the current ratio are likewise not meaningful for banks, whose balance sheets are not organized around a current/non-current distinction; regulators instead impose bank-specific capital and liquidity ratios, some introduced in the Basel III discussion of Chapter 3.

Capital-intensive industries such as utilities, airlines, and telecommunications typically carry substantial long-term debt and lower interest coverage than an asset-light firm such as Solstice Robotics, the comparison company of Section 4.4.1. Their relatively stable and predictable revenues (regulated utility rates, contracted

Wulin.Suo@Queensu.ca

infrastructure usage) let them service more debt safely than a firm with volatile cash flows. A debt-to-equity ratio of 1.5 might be unremarkable for a regulated utility but alarming for an unproven technology startup.

Asset-light service and technology businesses, more like Solstice than Meridian, often carry little debt, hold large cash balances, and derive most of their value from intangible assets (customer relationships, brand, intellectual property) that may barely appear on the balance sheet under conventional accounting. Market-based valuation multiples (Chapter 5) are therefore often used alongside, or instead of, book-value ratios for such firms, since the balance sheet may substantially understate the economic assets driving the business.

For any model built across many firms, ratios should be compared within a sector or peer group rather than against a universal threshold, and sector membership and capital intensity are often useful features to include directly in a predictive model.

This has a direct consequence for a distress score like Altman's Z′. A formula whose coefficients were estimated mainly on manufacturing and industrial firms will systematically misclassify even a perfectly healthy bank, whose debt-to-equity ratio would sit far above, and whose X_4 (book equity divided by total liabilities) far below, any threshold intended for an industrial company. Neither Meridian's Z′-score of 2.41 nor Solstice's 4.33, computed later in this chapter, is directly comparable to a Z′-score computed with the same coefficients for a bank or a capital-intensive utility. That is why sector-specific credit-scoring models, such as the bank-focused material in Chapter 9, are estimated on sector-specific data rather than reusing a general-purpose formula across every industry.

4.3.4 How Firms Create Value

Suppose a firm doubles its revenue and its balance sheet over five years, yet its shareholders end up no wealthier than if it had stayed the same size. Growth alone says nothing about value creation, because growth must be paid for with capital, and capital has a cost, whether lenders demand interest or shareholders demand a competitive return. A firm creates value only when it earns returns on invested capital above the cost of that capital; projects earning less than the required return destroy value however much bigger the firm becomes. This is one of the most important ideas in corporate finance, commonly expressed through economic profit,

$$\text{Economic profit} = \text{Invested capital} \times (\text{ROIC} - \text{WACC}), \tag{4.4}$$

where ROIC is the return on invested capital and WACC is the weighted average cost of capital. A simple value-creation diagram is shown in Figure 4.2.



Figure 4.2: A firm generates value when its projects earn a return above the cost of capital

This idea links financial statements to valuation. A firm that earns high returns on capital and reinvests profitably compounds value over time; one that earns low returns and uses excessive leverage may look large but create little wealth. A predictive model may forecast changes in revenue, but an analyst must still ask whether those changes translate into durable value creation.

Growth is therefore valuable when tied to profitable reinvestment and harmful when it consumes large amounts of capital with weak expected returns. Meridian grew its balance sheet from \$112.0 million to \$129.0 million, a 15% increase, while free cash flow turned negative. Whether this growth creates value depends on whether the additional capital will earn returns above its cost.

The statements cannot supply a cost of capital, but they supply everything else this calculation needs. Meridian's invested capital, the sum of its interest-bearing debt and equity, is \$58.0 + \$54.0 = \$112.0 million (Table 4.1), and its after-tax operating profit, or net operating profit after tax (NOPAT), is EBIT × (1 − tax rate) = 16.0 × 0.75 = \$12.0 million. This gives

$$\text{ROIC} = \frac{\text{NOPAT}}{\text{Invested capital}} = \frac{12.0}{112.0} \approx 10.7\%.$$

Wulin.Suo@Queensu.ca

If Meridian's weighted average cost of capital, estimated using the methods of Chapter 5, were 8% (close to the 8.08% that Chapter 5 estimates from Meridian's own statements), its economic profit would be $112.0 \times (0.107 - 0.08) \approx \$3.0$ million: modestly positive, so Meridian creates a small amount of value above its cost of capital despite its cash-flow strain. The distinction is initially counterintuitive. A firm can be a (marginal) value creator by this measure while burning cash and increasing leverage, because ROIC versus WACC asks whether capital is deployed productively, while free cash flow asks whether the firm is generating or consuming cash in the short run. A machine learning model trained to predict "financial health" from statement data must be clear about which of these two questions it is answering.

4.3.5 Common Financial Ratios and What They Reveal

Financial ratios convert raw statements into interpretable measures that let analysts compare firms, track trends, and identify concerns. They fall into profitability, liquidity, efficiency, solvency, and valuation categories. Table 4.4 collects the main ratio families together with their formulas and Meridian's Year 2 values.

Ratio	Formula	Meridian Value
Current ratio	Current assets / Current liabilities	$63.0/25.0 = 2.52$
Quick ratio	(Current assets − Inventory) / Current liabilities	$33.0/25.0 = 1.32$
Gross margin	Gross profit / Revenue	$54.0/150.0 = 36.0\%$
Operating margin	EBIT / Revenue	$16.0/150.0 = 10.7\%$
Net margin	Net income / Revenue	$9.0/150.0 = 6.0\%$
Debt-to-equity	Total debt / Equity	$58.0/54.0 = 1.07$
Interest coverage	EBIT / Interest expense	$16.0/4.0 = 4.0$
Asset turnover	Revenue / Total assets	$150.0/129.0 = 1.16$
Return on equity (ROE)	Net income / Equity	$9.0/54.0 = 16.7\%$

Table 4.4: Summary of common financial ratios, applied to Meridian, Year 2

The current ratio of 2.52 and quick ratio of 1.32 suggest that, on paper, Meridian has ample short-term resources to cover its current liabilities. The cash flow analysis adds nuance: much of the current assets is tied up in slow-moving inventory and receivables, and the cash balance fell during the year. A ratio from a single balance sheet date can look comfortable while masking a deteriorating cash trajectory, so ratios are more informative read together with the cash flow statement and their own trend.

A particularly useful decomposition is the DuPont identity, which breaks return on equity into three economically distinct drivers:

$$\text{ROE} = \underbrace{\frac{\text{Net income}}{\text{Revenue}}}_{\text{Net margin}} \times \underbrace{\frac{\text{Revenue}}{\text{Total assets}}}_{\text{Asset turnover}} \times \underbrace{\frac{\text{Total assets}}{\text{Equity}}}_{\text{Equity multiplier}}. \tag{4.5}$$

For Meridian, Net margin = 6.0%, Asset turnover = 1.16, and the equity multiplier is $129.0/54.0 = 2.39$. Multiplying these together at full precision, $0.060 \times 1.1628 \times 2.3889 \approx 16.67\%$, reproduces the directly computed ROE of 16.7%, as it must, since revenue and total assets cancel in the product; multiplying the rounded factors would open a small rounding gap, the same cascade as in the cash conversion cycle. The decomposition is diagnostic: Meridian's ROE is driven substantially by leverage (an equity multiplier above 2) rather than by high margins or asset efficiency. Two firms can report the same ROE for very different, and not equally sustainable, reasons, and the DuPont breakdown is the standard tool for telling them apart.

The three-factor version bundles taxes, interest, and core operating profitability into a single "net margin" term, hiding whether a margin is driven by operations or by a favorable tax or financing outcome. The five-factor extended DuPont identity splits net margin further,

$$\text{ROE} = \underbrace{\frac{\text{Net income}}{\text{Pretax income}}}_{\text{Tax burden}} \times \underbrace{\frac{\text{Pretax income}}{\text{EBIT}}}_{\text{Interest burden}} \times \underbrace{\frac{\text{EBIT}}{\text{Revenue}}}_{\text{Operating margin}} \times \text{Asset turnover} \times \text{Equity multiplier},$$

Wulin.Suo@Queensu.ca

isolating how much of operating profit survives interest expense and taxation. For Meridian, the tax burden is $9.0/12.0 = 0.75$ (the firm retains 75% of pretax income after taxes, consistent with the 25% tax rate used throughout this chapter), the interest burden is $12.0/16.0 = 0.75$ (the firm retains 75% of EBIT after interest expense), and the operating margin is $16.0/150.0 \approx 10.7\%$, as computed in Section 4.3.5. Multiplying all five factors,

$$0.75 \times 0.75 \times 0.107 \times 1.163 \times 2.389 \approx 16.7\%,$$

matches the directly computed ROE. The extra split is diagnostic. Meridian's two burden ratios are both 0.75, so taxes and interest erode profitability in the same proportion. A firm with a much lower interest burden (heavily levered, paying substantial interest) and a higher tax burden could show the same three-factor net margin as Meridian with a completely different cost structure, a difference only the five-factor decomposition surfaces.

Liquidity ratios such as the current ratio and quick ratio assess how easily a firm can meet short-term obligations. Efficiency ratios such as inventory turnover and receivables turnover show how quickly it converts resources into cash. Solvency ratios such as debt-to-equity and interest coverage show how vulnerable it is to distress. Valuation ratios such as price-to-earnings and price-to-book compare market expectations with fundamental performance, though they need market prices beyond the statements.

Ratios depend on context. A high leverage ratio may suit a stable utility but endanger a startup with volatile cash flows. Low inventory turnover may indicate weak sales or poor inventory management, or in some sectors a strategy of high service levels. Ratios are starting points for a conversation about the business, not final judgments.

4.4 Comparison and Common-Size Analysis

A ratio means little in isolation. This part introduces a second company and the common-size representation that makes two firms of very different scale directly comparable.

4.4.1 A Second Company for Comparison: Solstice Robotics Inc.

Ratios are most informative compared across firms or across time. Consider a second fictional company, Solstice Robotics Inc., a faster-growing, less capital-intensive firm, with the Year 2 figures in Table 4.5.

Cash and equivalents	40.0
Accounts receivable	20.0
Inventory	10.0
Total current assets	**70.0**
Property, plant, and equipment (net)	30.0
Total assets	**100.0**
Accounts payable	15.0
Long-term debt	10.0
Total liabilities	**25.0**
Shareholders' equity	75.0
Revenue	120.0
Cost of goods sold	48.0
SG&A	40.0
Depreciation and amortization	5.0
Interest expense	1.0
Net income	19.5

Table 4.5: Solstice Robotics Inc. – Summary Financials, Year 2 ($ millions)

Table 4.6 places Meridian and Solstice side by side on the same ratios computed in Table 4.4.

Wulin.Suo@Queensu.ca

Ratio	Meridian	Solstice
Current ratio	2.52	4.67
Quick ratio	1.32	4.00
Gross margin	36.0%	60.0%
Operating margin	10.7%	22.5%
Net margin	6.0%	16.25%
Debt-to-equity	1.07	0.13
Interest coverage	4.0	27.0
Return on equity	16.7%	26.0%

Table 4.6: Meridian versus Solstice, Year 2

Solstice's ROE of 26% is higher than Meridian's 16.7%, but the DuPont identity from Section 4.3.5 reveals that the two firms get there in opposite ways. Applying the same decomposition to Solstice: net margin 16.25%, asset turnover $120.0/100.0 = 1.20$, and equity multiplier $100.0/75.0 = 1.33$, so $0.1625 \times 1.20 \times 1.33 \approx 26.0\%$, matching directly.

Solstice's high ROE is driven almost entirely by superior margins (16.25% net margin versus Meridian's 6.0%), with very little help from leverage (an equity multiplier of 1.33 versus Meridian's 2.39). Meridian, by contrast, achieves a respectable but lower ROE mostly *because* of leverage. An analyst, or a machine learning model trained naively on ROE alone, would rank the two firms similarly although their financial structures, and therefore their risk profiles, are almost opposite. The DuPont decomposition exists to surface what a single summary ratio hides.

4.4.2 Common-Size Statements

A second way to make firms comparable is the common-size statement, in which every income statement line is expressed as a percentage of revenue and every balance sheet line as a percentage of total assets. Table 4.7 shows Meridian's and Solstice's income statements this way.

	Meridian	Solstice
Cost of goods sold	64.0%	40.0%
Gross profit	36.0%	60.0%
SG&A	20.0%	33.3%
Depreciation and amortization	5.3%	4.2%
Operating income (EBIT)	10.7%	22.5%
Interest expense	2.7%	0.8%
Net income	6.0%	16.25%

Table 4.7: Common-size income statements, Year 2 (percent of revenue)

Common-size statements make cost-structure differences visible in a way raw dollar figures do not: Solstice spends proportionally much less on cost of goods sold (40% of revenue versus Meridian's 64%) but more on SG&A (33.3% versus 20%), consistent with a less capital-intensive, more sales-and-marketing-driven business. Expressing every line relative to a common base is also the kind of feature engineering that makes statement data usable as input to the machine learning models of Chapter 6, since raw dollar figures are not comparable across firms of different sizes.

Wulin.Suo@Queensu.ca

The same technique applies to the balance sheet, expressing every line as a percentage of total assets rather than of revenue. For Meridian's Year 2 balance sheet, this gives

$$\begin{aligned} \text{Cash} &= 6.98\%, & \text{Accounts payable} &= 13.18\%, \\ \text{Receivables} &= 18.60\%, & \text{Short-term debt} &= 6.20\%, \\ \text{Inventory} &= 23.26\%, & \text{Long-term debt} &= 38.76\%, \\ \text{Property, plant, and equipment} &= 51.16\%, & \text{Equity} &= 41.86\%, \end{aligned}$$

of total assets. More than half of Meridian's balance sheet, 51.16%, is tied up in fixed, illiquid property, plant, and equipment, while liabilities and equity are split 58.14% to 41.86%, a far more leveraged structure than Solstice's, whose debt-to-equity ratio is only 0.13 (Table 4.6). Common-size balance sheets reveal such structural differences at a glance, before any explicit ratio is computed.

Ratios can also be combined into a single index that predicts financial distress, rather than read one at a time as in Table 4.4. The best known is Altman's Z-score. Four of its five components are variations on ratios already computed: a liquidity measure related to the working capital of Section 4.3.1, a leverage cushion related to the debt-to-equity ratio of Section 4.3.2, and earning-power and asset-turnover measures related to the DuPont components of Section 4.3.5. The fifth measures cumulative retained profitability, and the weights are coefficients Altman estimated statistically from historical bankruptcy data rather than derived from theory. For a private (non-publicly-traded) firm like Meridian, the Z′-score variant is

$$Z' = 0.717X_1 + 0.847X_2 + 3.107X_3 + 0.420X_4 + 0.998X_5, \tag{4.6}$$

where

$$\begin{aligned} X_1 &= \text{Working capital/Total assets}, & X_2 &= \text{Retained earnings/Total assets}, \\ X_3 &= \text{EBIT/Total assets}, & X_4 &= \text{Book value of equity/Total liabilities}, \\ X_5 &= \text{Sales/Total assets}. \end{aligned}$$

Assuming, for illustration, that all of Meridian's \$54.0 million in equity represents retained earnings, the components are

$$X_1 = \frac{38.0}{129.0} = 0.295, \quad X_2 = \frac{54.0}{129.0} = 0.419, \quad X_3 = \frac{16.0}{129.0} = 0.124,$$

$$X_4 = \frac{54.0}{75.0} = 0.720, \quad X_5 = \frac{150.0}{129.0} = 1.163,$$

giving

$$Z' = 0.717(0.295) + 0.847(0.419) + 3.107(0.124) + 0.420(0.720) + 0.998(1.163) \approx 2.41.$$

Altman's original thresholds classify $Z' > 2.9$ as a "safe" zone, $1.23 < Z' < 2.9$ as a "gray" zone of ambiguous risk, and $Z' < 1.23$ as a "distress" zone with a high historical likelihood of bankruptcy within two years. Meridian's score of 2.41 falls in the gray zone, consistent with everything the chapter has found: healthy-looking ratios on the surface, a deteriorating cash position and rising leverage underneath. The score is useful because it compresses many individual signals into one number, but like any single ratio it is a starting point for investigation, not a verdict. Altman's coefficients were estimated with statistical discriminant analysis, a direct ancestor of the classification methods in Chapter 6.

The same score for Solstice Robotics (Section 4.4.1) makes the contrast concrete. Using Solstice's Year 2 figures, and again assuming all of Solstice's \$75.0 million in equity represents retained earnings, working capital is $70.0 - 15.0 = 55.0$ (current assets less accounts payable, its only current liability), so

$$X_1 = \frac{55.0}{100.0} = 0.550, \quad X_2 = \frac{75.0}{100.0} = 0.750, \quad X_3 = \frac{27.0}{100.0} = 0.270,$$

$$X_4 = \frac{75.0}{25.0} = 3.000, \quad X_5 = \frac{120.0}{100.0} = 1.200,$$

Wulin.Suo@Queensu.ca

where Solstice's EBIT of \$27.0 million matches the interest coverage figure in Table 4.6 ($27.0 \times \$1.0$ million interest expense = \$27.0 million), giving

$$Z' = 0.717(0.550) + 0.847(0.750) + 3.107(0.270) + 0.420(3.000) + 0.998(1.200) \approx 4.33.$$

Solstice's score of 4.33 sits comfortably in Altman's "safe" zone, in contrast to Meridian's 2.41, for reasons the chapter has already surfaced: Solstice carries far less leverage ($X_4 = 3.00$ versus Meridian's 0.72) and a far larger cushion of retained earnings and working capital relative to its assets. The Z-scores summarize quantitatively what the DuPont decomposition and Table 4.6 showed: firms reach their profitability through different combinations of margin, efficiency, and leverage, and a distress score is only as informative as the analyst's understanding of which component drives it.

4.5 From Statements to Forecasts

Statements describe the past; valuation needs the future. This part connects reported figures to the cash flows Chapter 5 discounts, notes how far reported numbers can be trusted, and builds a forecast from them.

4.5.1 Connecting Financial Statements to Valuation

Financial statement analysis often serves valuation. Investors value a firm by estimating the cash flows it can generate and discounting them, and the statements help identify those cash flows: revenues, margins, working capital needs, investment requirements, and financing choices all affect them.

A simple valuation logic is that a firm's value is the present value of future free cash flows,

$$V = \sum_{t=1}^{T} \frac{FCF_t}{(1+r)^t},$$

where r reflects the riskiness of those cash flows. The statements supply the raw material: revenue growth and margin trends inform the profit forecast, balance sheet data inform working capital and investment needs, and the cash flow statement shows how earnings have historically converted into cash. Projecting Meridian's free cash flow forward on its current negative trajectory would imply a far less attractive valuation than net income alone suggests, which is why practitioners insist on free cash flow rather than accounting earnings as the input to valuation.

Free cash flow, however, comes in two varieties that answer two different questions, and conflating them is a common and consequential error. Both start from CFO, the operating cash flow of Section 4.2.3. Free cash flow to the firm (FCFF) measures cash available to *all* capital providers, debt and equity holders combined, before any financing decisions,

$$\text{FCFF} = \text{CFO} + \text{Interest expense} \times (1 - \text{tax rate}) - \text{Capex}, \tag{4.7}$$

adding back after-tax interest because it is a payment *to* one class of capital provider (debtholders), not a cost of the business itself. Free cash flow to equity (FCFE) measures cash available to equity holders after debtholders have been paid and any net new borrowing received,

$$\text{FCFE} = \text{CFO} - \text{Capex} + \text{Net borrowing}. \tag{4.8}$$

Applying both to Meridian's Year 2 statements (CFO = \$6.0 million, Capex = \$14.0 million, interest expense \$4.0 million, tax rate 25%, net new debt issued \$12.0 million),

$$\text{FCFF} = 6.0 + 4.0(1 - 0.25) - 14.0 = -\$5.0 \text{ million}, \qquad \text{FCFE} = 6.0 - 14.0 + 12.0 = \$4.0 \text{ million}.$$

The two measures have opposite signs. Meridian's business, independent of how it is financed, consumed \$5.0 million more cash than it generated, but \$12.0 million of new debt more than covered that gap, leaving \$4.0 million for equity holders. An analyst who computed only FCFE would see a healthy positive number and could miss that the core operations are cash-negative and that the positive figure exists only because of a financing decision, an increase in leverage, that carries risks this chapter has already flagged. FCFF is the

Wulin.Suo@Queensu.ca

natural input to an enterprise-value calculation discounted at the WACC of Chapter 5, since it is the cash available to every capital provider WACC blends together. FCFE is the input to a direct equity valuation discounted at the cost of equity alone. Pairing the wrong rate with the wrong cash flow (WACC with FCFE, or the cost of equity with FCFF) produces an error that does not announce itself, only a wrong number.

For AI applications, a machine learning system can find patterns in balance sheet or cash flow data, but those patterns must be interpreted in economic terms. A distress model should be grounded in real indicators such as falling liquidity, rising leverage, declining margins, and weak cash generation; otherwise it may capture noise rather than economic reality.

4.5.2 Accounting Quality and the Limits of Reported Numbers

Reported numbers are not always fully transparent. Accounting rules allow management some discretion, particularly in areas such as revenue recognition, depreciation, reserves, and provisioning. A firm may use aggressive accounting policies to smooth earnings or to present a stronger balance sheet than the underlying reality supports.

Careful analysts therefore ask how the statements were produced, not only what they say. They look for unusual changes in reserves, one-time charges, deferred revenue patterns, and recurring non-cash items, and compare reported figures with cash flow trends and outside information such as industry reports, competitor behavior, or macroeconomic indicators. Meridian's widening gap between net income and operating cash flow, driven by growing receivables and inventory, is a textbook signal to investigate: is the firm's revenue recognition aggressive, are customers slowing their payments, or is inventory building up because of weakening demand?

This gap between reported earnings and cash generation can itself be quantified. One simple accruals ratio measure is

$$\text{Accruals ratio} = \frac{\text{Net income} - \text{Cash flow from operations}}{\text{Total assets}}.$$

For Meridian, this is $(9.0 - 6.0)/129.0 \approx 2.3\%$: net income exceeds operating cash flow by an amount equal to 2.3% of total assets. Research on earnings quality has repeatedly found that firms with unusually high accruals ratios, whose reported earnings substantially exceed cash generation, tend on average to see earnings growth disappoint and stock returns underperform afterward. A large gap between earnings and cash warns that current profitability may be inflated by aggressive accounting, temporary working-capital effects, or both, rather than durable operating improvement. A single year's accruals ratio is weak evidence on its own. The signal becomes more informative tracked over several years and compared with the firm's own history and its peers, a sector-adjusted, longitudinal comparison that a cross-sectional machine learning model of the kind in Chapter 6 is well suited to automate.

Case Vignette: Wirecard and the Cash That Was Not There

Wirecard, a German payments processor that had joined the DAX index of Germany's largest listed companies in 2018, is the cautionary case for anyone who trusts the accruals ratio alone. On 18 June 2020 its auditor, EY, declined to sign off on the 2019 accounts because it could not confirm that €1.9 billion of cash, about a quarter of the balance sheet, sat in trustee accounts at two banks in the Philippines. Both banks said Wirecard was not their client, and the documents purporting to show the balances were forgeries. On 22 June Wirecard said the money most likely did not exist, and on 25 June it filed for insolvency. A KPMG special audit published that April had already been unable to verify the business with third-party acquiring partners that Wirecard reported as about half of its revenue and operating profit, and Financial Times reporting since early 2019 had argued that those partners and the cash attributed to them were not what they seemed (McCrum 2022). The earnings were real on paper; the cash behind them was claimed to be held by outside trustees, out of the reach of anyone who wanted to count it.

The stylized numbers below, for a fictional processor, Halcyon Payments (€ millions), show why this chapter's ratios flag such a scheme at one stage and are defeated at the next, and which simple extra check restores them. In Year 2, Halcyon's revenue grows 40% but receivables almost triple, so days sales outstanding,

Wulin.Suo@Queensu.ca

the receivables leg of the cash conversion cycle, doubles; operating cash flow falls even as profit rises.

	Year 1	Year 2	Year 3
Revenue	1,000	1,400	1,900
Net income	150	240	340
CFO (reported)	210	160	430
of which received into trustee accounts	0	0	500
Receivables	150	420	300
Cash at own banks / at trustees	400/0	450/0	380/500
Total assets	2,000	2,600	3,400
DSO (days)	54.8	109.5	57.6
CFO / EBITDA	0.84	0.41	0.80
Accruals ratio, reported	−3.00%	3.08%	−2.65%
Accruals ratio, own accounts	−3.00%	3.08%	12.06%

(EBITDA is 250, 390, and 540.) Year 2 is the pattern Meridian's analysis taught an analyst to flag: the accruals ratio turns positive at $(240 - 160)/2{,}600 = 3.08\%$ and cash conversion, CFO divided by EBITDA, falls from 0.84 to 0.41. In Year 3 the flags go quiet. The receivables are reported as collected, DSO returns to 57.6 days, cash conversion recovers to 0.80, and the accruals ratio falls to $(340 - 430)/3{,}400 = -2.65\%$, better than Year 1. But €500 million of that operating cash flow went into accounts held by third-party trustees, not into Halcyon's own banks, whose balance fell by €70 million. Recomputed on cash the firm itself controls, operating cash flow is $430 - 500 = -70$ million euros and the accruals ratio is $(340 + 70)/3{,}400 = 12.06\%$, the worst of the three years; over Years 2 and 3 together the firm reported €580 million of profit and brought €90 million of operating cash into its own accounts. The trustee balance is 56.8% of the firm's cash and 14.7% of its total assets.

Two lessons follow. A ratio is only as good as the line items under it: accruals and cash-conversion screens test whether earnings became cash, and a scheme that fabricates the cash, not just the earnings, passes them. And the defense is not a cleverer ratio but a question about existence and control: where is the cash, who holds it, and has anyone independent confirmed it directly with the bank, the step whose omission for years was at the heart of the criticism of Wirecard's audits. A machine learning model trained on reported figures inherits every assumption the statements make, including the assumption that the cash on the balance sheet exists.

Such a model may also learn patterns that are partly artifacts of accounting conventions rather than economic reality, so the analyst must understand the reporting environment and remain skeptical of simplistic interpretations.

4.5.3 Forecasting from Statements

Statement analysis is especially valuable for forecasting. Analysts project revenues, costs, margins, capital expenditures, and working capital needs, which then feed valuation models; a well-built forecast shows whether a firm is likely to create value, consume cash, or need external financing.

A simple forecasting approach ties each balance sheet and income statement line to revenue. For example, if COGS is historically 64% of revenue (as it is for Meridian: $96.0/150.0 = 64\%$), and receivables are historically 16% of revenue (24.0/150.0), then a revenue forecast for Year 3 immediately implies forecasts for COGS and receivables under a "percent of sales" model. The method is intentionally simple and can be extended with separate assumptions for margins, working capital efficiency, and capital intensity as information arrives.

Table 4.8 carries the method through, projecting Meridian's full Year 3 income statement and key working-capital lines under two assumptions: 8% revenue growth, and every other line holding its historical percent-of-revenue (or, for accounts payable, percent-of-COGS) ratio constant, with interest expense held fixed at its Year 2 dollar level since it depends on the debt outstanding rather than on sales.

Two things stand out from this forecast. First, net income is projected to grow from $9.0 million to $9.96 million, a 10.7% increase, faster than the 8% revenue growth assumption, because fixed interest expense does not grow with revenue, so a larger share of incremental operating income flows through to the bottom line, an instance of financial leverage. Second, accounts receivable and inventory are forecast to grow with revenue, so, as in the Year 2 cash flow statement, this growth consumes cash even when the business performs

Wulin.Suo@Queensu.ca

Line	Year 2	Year 3 (forecast)	Basis
Revenue	150.0	162.0	8% growth assumption
Cost of goods sold	96.0	103.68	64.0% of revenue
Gross profit	54.0	58.32	
SG&A	30.0	32.40	20.0% of revenue
Depreciation and amortization	8.0	8.64	5.3% of revenue
Operating income (EBIT)	16.0	17.28	
Interest expense	4.0	4.0	held constant
Pretax income	12.0	13.28	
Taxes (25%)	3.0	3.32	
Net income	9.0	9.96	
Accounts receivable	24.0	25.92	16.0% of revenue
Inventory	30.0	32.40	20.0% of revenue
Accounts payable	17.0	18.36	17.7% of COGS

Table 4.8: A percent-of-sales forecast for Meridian, Year 3 ($ millions)

as historically expected. A complete forecast carries the projected balance sheet into a projected cash flow statement to check whether Meridian's growth is self-funding or still needs external financing, the free-cash-flow valuation exercise of Section 4.5.1. Forecasting from statements is therefore not mechanical: the percent-of-sales method makes a specific, checkable assumption (that these ratios remain constant), and the exercise is worth as much for questioning that assumption as for computing its consequences.

That complete forecast resolves a question left open since Table 4.3. Year 3 cash flow from operations is projected net income plus depreciation, less the increase in receivables and inventory, plus the increase in payables:

$$\text{CFO}_3 = 9.96 + 8.64 - (25.92 - 24.0) - (32.40 - 30.0) + (18.36 - 17.0) = 15.64,$$

more than double Year 2's $6.0 million, because the dollar growth in net income and depreciation outpaces the dollar growth in working capital even though every underlying ratio is held constant. Holding capital expenditures at their Year 2 percentage of revenue ($14.0/150.0 = 9.33\%$) gives a Year 3 projected capex of $0.0933 \times 162.0 \approx \$15.12$ million, so

$$\text{FCF}_3 = \text{CFO}_3 - \text{Capex}_3 = 15.64 - 15.12 = \$0.52 \text{ million},$$

compared to Year 2's free cash flow of $6.0 - 14.0 = -\$8.0$ million. Under these assumptions, Meridian's free cash flow turns from sharply negative to barely positive with no change in margins or efficiency, because operating cash flow scales faster with revenue than working-capital investment once the heaviest receivables-and-inventory buildup is past. This $0.52 million is the Year 3 figure with which Chapter 5's discounted-cash-flow valuation of Meridian begins; an analyst extrapolating Year 2's deeply negative free cash flow would undervalue Meridian relative to what the same assumptions imply.

Forecasting balances structure and judgment. Some assumptions rest on historical averages, others on strategy, industry conditions, and the macroeconomic outlook. Economies of scale may be expected to improve margins, but competition can delay the improvement; planned capacity expansion can suffer delays and cost overruns. Financial forecasting is both quantitative and interpretive.

The best forecasts usually rest on simple, transparent assumptions that can be tested and updated. An overly complex model may look precise while offering little economic insight, a lesson that applies equally to financial analysis and machine learning.

4.6 Challenges in Financial Statement Analysis

Several practical challenges complicate the use of financial statements, especially as inputs to quantitative or machine learning models.

Comparability across firms and time. Accounting standards differ across countries (for example, the International Financial Reporting Standards, or IFRS, used across much of the world, versus U.S. Generally

Wulin.Suo@Queensu.ca

Accepted Accounting Principles, or GAAP), and even within a single standard, firms have discretion over methods such as inventory costing (FIFO versus LIFO) or depreciation schedules. A model trained on ratios pooled across many firms may compare numbers that are not comparable unless these differences are adjusted for.

Reporting lag and low frequency. Financial statements are published quarterly or annually, weeks after the period ends. They are a slow-moving input compared with market prices or transaction data, and a model relying mainly on them reacts to changes in a firm's condition only after a considerable delay.

Restatements and revisions. Reported figures are sometimes restated after publication, because of errors, changes in accounting policy, or audit corrections. A dataset built from "as first reported" figures can differ meaningfully from one incorporating later restatements, and using the wrong vintage can introduce a subtle look-ahead bias into a backtest.

Manipulation and earnings management. Management has some latitude in how figures are presented. Detecting earnings management, for example through unusual accruals, is an active area of quantitative research, and a model that ignores this risk may be fooled by cosmetically strong statements.

Non-financial and off-balance-sheet items. Important sources of risk or value, such as operating leases (in some reporting regimes), pending litigation, environmental liabilities, or human capital, may not appear cleanly on the balance sheet, so statement-based models can miss economically material risks.

Sector heterogeneity. A ratio that signals distress in one industry may be normal in another; utilities and airlines routinely carry leverage that would alarm a software company. A cross-sectional model must account for sector context rather than apply a single threshold.

None of this means abandoning statement-based analysis. Financial statements are a valuable but imperfect and delayed signal, to be combined with market data, qualitative judgment, and, where available, higher-frequency alternative data.

4.7 Key Takeaways

Financial statements are the backbone of corporate finance. The balance sheet shows the firm's financial structure, the income statement its performance, and the cash flow statement the movement of cash; together they show whether a business is profitable, liquid, solvent, and capable of creating value.

The Meridian example showed a firm reporting solid net income and comfortable liquidity ratios while burning cash and increasing leverage, a discrepancy visible only when all three statements and their ratios are examined together. Understanding the statements comes before advanced forecasting or valuation, and before trusting any machine learning model built on accounting data. The Wirecard case marked the limit of that skepticism: ratios test whether earnings turned into cash, so a fraud that fabricates the cash passes them, and only asking who holds the cash, and whether anyone independent has confirmed it, catches it.

The comparison with Solstice Robotics reinforced a point that recurs throughout the book: a single summary statistic, whether a ratio, a Z'-score, or a model's output score, can mask very different underlying stories, which the DuPont identity and common-size statements exist to unpack.

The sector-specific discussion cautions against a single cross-firm model that ignores how differently a bank, a utility, and a technology company are financed, even though all report the same three statements.

Chapter 5's valuation methods take as input cash flows that come from the statement analysis, and the skepticism about statement quality, developed here. Chapter 9's credit models build on this chapter's leverage, liquidity, and distress-scoring ideas, with statistical estimation in place of a fixed formula like Altman's Z'-score.

4.8 Suggested Exercises

4.1 Explain the difference between assets, liabilities, and equity, and verify that the accounting identity holds for both years of Meridian's balance sheet.

4.2 Using Table 4.1 and Table 4.2, compute Meridian's current ratio, quick ratio, and debt-to-equity ratio for Year 1, and compare them to the Year 2 values computed in the chapter.

4.3 Describe why cash flow can differ from accounting profit, using Meridian's Year 2 numbers as a concrete illustration.

Wulin.Suo@Queensu.ca

4.4 Compute Meridian's free cash flow if capital expenditures had instead been \$6.0 million rather than \$14.0 million, holding all other cash flow items fixed, and discuss how this would change your assessment of the firm.

4.5 Using the DuPont identity, decompose Meridian's ROE for Year 1 (assume net income of \$7.5 million and revenue of \$135.0 million for Year 1) and compare the drivers of ROE across the two years.

4.6 Explain why a company with growing sales, like Meridian, might still face financial distress, referencing the cash conversion cycle.

4.7 Discuss two ways that accounting discretion could distort a machine learning model trained to predict corporate default using financial ratios.

4.8 Propose a percent-of-sales forecast for Meridian's Year 3 revenue, COGS, and accounts receivable, assuming 8% revenue growth and that historical ratios to revenue remain constant.

4.9 Using Table 4.5, verify the current ratio, quick ratio, and debt-to-equity ratio reported for Solstice in Table 4.6.

4.10 Explain, using the DuPont identity, why Meridian and Solstice can have different return on equity even though the gap between their net margins (6.0% versus 16.25%) is larger than the gap in their ROE (16.7% versus 26.0%).

4.11 Using Table 4.5, construct a common-size balance sheet (each line as a percent of total assets) for Solstice's Year 2 balance sheet, and compare it to Meridian's common-size balance sheet computed in this chapter, identifying the single largest structural difference between the two firms.

4.12 Compute Meridian's accruals ratio for a hypothetical Year 3 in which net income is \$11.0 million and cash flow from operations is \$4.0 million (total assets unchanged at \$129.0 million), and explain whether this year's figure is more or less concerning than Year 2's.

4.13 Using the Z′-score formula, recompute Meridian's distress score assuming Year 1 figures instead of Year 2 (Year 1 EBIT can be assumed at \$13.0 million, retained earnings equal to Year 1 equity of \$52.0 million, and other Year 1 figures from Table 4.1), and discuss whether Meridian's financial health appears to be improving or deteriorating between the two years.

4.14 Verify the Solstice Z′-score computed in Section 4.4.1 by recomputing X_1 through X_5 from Table 4.5, and explain in one sentence which single component contributes most to the gap between Solstice's score and Meridian's.

4.15 Using Table 4.8's Year 3 projections, recompute Meridian's projected free cash flow if the revenue growth assumption were 12% instead of 8% (holding every ratio in Table 4.8 fixed at its stated percentage of revenue or COGS), and explain why free cash flow does not simply scale in proportion to revenue growth.

4.16 Compute Meridian's return on invested capital and economic profit if its weighted average cost of capital were 12% instead of 8%, holding NOPAT and invested capital at their Year 2 values, and explain what this implies about whether Meridian is creating or destroying value at that higher cost of capital.

4.17 Rework the Halcyon Payments case of Section 4.5.2 for a milder Year 3, in which only €250 million of operating cash flow goes into trustee accounts, so that own-bank cash ends the year at €630 million (reported CFO of €430 million, net income €340 million, and total assets €3,400 million unchanged). Compute operating cash flow into the firm's own accounts, the reported and own-account accruals ratios, and the trustee share of total cash. Which of the chapter's screens would flag this year, and what single question to the auditor would settle the matter?

4.18 **Challenge (real data).** Pick any US public company and download its most recent 10-K balance sheet and income statement figures directly from the SEC EDGAR XBRL `companyfacts` API (free, no login required). (a) Compute the same ratio suite Section 4.3.5 computed for Meridian: current ratio, quick ratio, debt-to-equity, and a three-factor DuPont ROE decomposition. (b) Using this chapter's Altman Z′-score formula, classify the firm into the safe, gray, or distress zone, and comment on whether the classification matches your own intuition about the company's financial health. (c) Real XBRL filings tag the same economic concept (for example, inventory or long-term debt) under

Wulin.Suo@Queensu.ca

dozens of different standardized and company-specific tags across firms and years; describe at least one tag-matching or missing-data problem you encountered while pulling the figures needed for part (a), and how you resolved it.

4.9 Further Reading

- Brealey et al. (2025), *Principles of Corporate Finance*. Covers the corporate-finance logic behind cash flow, capital structure, and value creation that underlies this chapter's ratio analysis and forecasting material.
- Penman (2013), *Financial Statement Analysis and Security Valuation*. A rigorous treatment of how ratio analysis and common-size statements, introduced in this chapter with Meridian and Solstice, connect to security valuation more broadly.
- Damodaran (2012), *Investment Valuation*. Extends this chapter's percent-of-sales forecasting into the full range of valuation techniques developed formally in Chapter 5.
- Palepu et al. (n.d.), *Business Analysis and Valuation*. A case-based approach to interpreting financial statements and diagnosing accounting quality issues, complementing this chapter's Altman Z-score and distress-analysis material.
- McCrum (2022), *Money Men: A Hot Startup, a Billion-Dollar Fraud, a Fight for the Truth*. The Financial Times journalist whose reporting exposed Wirecard tells the story from the inside, a readable account of how strong reported numbers, an unquestioned audit, and a defensive regulator kept a fabricated business alive for years.

Wulin.Suo@Queensu.ca